column-hse-z: hydrostatic columns on a height grid

Physics of `run.mode = "column-hse-z"` (HSEMHS). 24 Oct 2026.

1. What the mode does

Given the temperature $T(x, y, z)$ of a solar atmosphere on a geometric-height grid, the mode computes the gas pressure P, the density ρ, the electron density and the optical depth τ_{500} that every vertical column would have if it were in **hydrostatic equilibrium on its own**. Each column is independent of its neighbours: no horizontal force balance is attempted and the magnetic field plays no role. The result is the non-magnetic product of the code, the reference against which the magnetic modes are measured, and the starting state of `mhs-poisson-z` and `full-mhs-z`.

2. Quantities

Symbol	Meaning	Unit
x, y, z	horizontal coordinates and height; z is given top first (strictly decreasing)	m
T	temperature, from the input, **never changed**	K
P, u	gas pressure; the code integrates $u = \ln(P/1\,\mathrm{Pa})$	Pa, –
ρ	mass density, $\rho = \rho(T, P)$ from the equation of state (EOS)	$\mathrm{kg\,m^{-3}}$
n_e, $\bar{\mu}$	electron density; mean molecular weight	$\mathrm{m^{-3}}$, –
g	surface gravity (274 for the Sun; from the file attribute `gravity_m_s2`)	$\mathrm{m\,s^{-2}}$
H	pressure scale height $P/(\rho g) = k_B T/(\bar{\mu} m_{\mathrm{H}} g)$	m
P_{ref}	pressure fixed at the anchor (bottom plane by default)	Pa
χ_{500}	extinction coefficient at 500 nm (absorption plus scattering)	$\mathrm{m^{-1}}$
τ_{500}	optical depth at 500 nm, counted downward from the top	–
m	column mass above a point	$\mathrm{kg\,m^{-2}}$

3. The equation

In a static, non-magnetic atmosphere the vertical force balance on a gas parcel is

$$\frac{dP}{dz} = -\rho g.$$

With $\rho = \rho(T, P)$ and $T(z)$ given, this is an ordinary differential equation along each column. Written for $u = \ln P$,

$$\boxed{\frac{du}{dz} = -\frac{g\,\rho\big(T(z), e^u\big)}{e^u} = -\frac{1}{H(T, P)}}$$

For an ideal gas of fixed composition and ionisation, H depends on T only and the solution is the familiar exponential atmosphere $P(z) = P_{\mathrm{ref}} \exp\big(-\int dz/H\big)$. In the Sun $\bar{\mu}$ changes with ionisation and molecule formation, so H depends on P too; that is why the equation is integrated numerically with a full EOS. In the photosphere $H \approx 100$–150 km; over the 3.5 Mm of the production box the pressure falls by about eight orders of magnitude (median $10^{5.0}$ Pa at the bottom, $10^{-3.1}$ Pa at the top, with a spread of 5.5 dex between columns there), which is why $u = \ln P$ rather than P is integrated.

The column mass is integrated along with u: $dm/dz = -\rho$, so that $P(z) - P_{\mathrm{top}} = g\,m(z)$ can be checked.

4. Where the pressure is fixed (the anchor)

A first-order equation needs one value per column. The default is a **bottom anchor**: the same pressure P_{ref} at the lowest grid level in every column, and integration upward. For the production input (`IO/input-mhs-hse.nc`, Bifrost `en024048_hion` snapshot 385 cropped to $-0.49 \ldots + 2.98$ Mm) $P_{\mathrm{ref}} = 1.035 \times 10^5$ Pa, the median Bifrost pressure on that plane (spread 0.08 dex). Alternatives are a top anchor (default 0.1 Pa) and an anchor at a chosen height z_{ref} (integrated both ways).

Why the bottom: the deep layers are where the pressure is best known and least sensitive to the details above. A top anchor makes the photospheric pressure depend on the full chromospheric and transition-region temperature structure, which differs strongly from column to column.

5. Equation of state and opacity

ρ, n_e and χ_{500} come from a table built with Wittmann's LTE equation of state (solar composition, H_2 included, a Saha branch for H and He above a switch temperature) and the continuum opacity at 5000 Å, interpolated bicubically in $(\ln T, \ln P)$. The table `cache/eos_table_fullz.h5` covers 2172 K–1.78 MK and 10^{-13}–10^{15} Pa. One EOS state supplies ρ, n_e, the electron pressure and χ_{500}, so they are consistent with each other.

6. Optical depth

With $P(z)$ and $T(z)$ known, χ_{500} is known at every level and

$$\tau_{500}(z) = \tau_{\rm top} + \int_z^{z_{\rm top}} \chi_{500}\, dz',$$

integrated downward with the trapezoid rule. $\tau_{\rm top}$ accounts for the gas above the box. The default closure (`exp_tail`) assumes that χ keeps falling exponentially with the scale height of the top two levels, $H_\chi = (z_1 - z_2)/\ln(\chi_2/\chi_1)$, giving $\tau_{\rm top} = \chi_1 H_\chi$. Where that is not meaningful (χ not falling, or $H_\chi > 5000$ km) half a cell, $\frac{1}{2}\chi_1(z_1 - z_2)$, is used and the column is flagged. The height of $\tau_{500} = 1$, $z_{\tau 1}(x, y)$, is the "visible surface": spectral lines form at fixed τ, so this is what an inversion sees.

7. What the result means, and what it does not

- Each column is exactly hydrostatic. **Horizontally the atmosphere is not in balance**: neighbouring columns with different temperatures have different $P(z)$, so there are horizontal pressure gradients that nothing balances. The mode reports them as the horizontal force residual R_x, R_y (with $B = 0$), and labels the output `vertical_hse_only`.
- A non-magnetic atmosphere can be in full 3-D static balance only if it is horizontally uniform (the curl of the force balance requires $\partial_x \rho = \partial_y \rho = 0$). A structured atmosphere in true 3-D balance needs the Lorentz force, which is what the `mhs-*` and `full-mhs-z` modes add.
- The temperature is never altered.

8. Typical result

Production input, full cube ($504 \times 504 \times 177$): $z_{\tau 1}$ has percentiles 1/50/99 of −29 / 64 / 127 km. The column-to-column scatter of $z_{\tau 1}$ is set entirely by the temperature structure, because every column carries the same bottom pressure.